

PORTFOLIO PROJECT DECK

ERP Manufacturing Data Analytics & Pipeline Optimization

End-to-End Data Cleaning Pipeline in SQL Server & Interactive Quality Analytics in Power BI

Role: Lead Data Analyst / Manufacturing Analytics Specialist

Tech Stack: MS SQL Server, Power BI, DAX, Star Schema Modeling

Domain: MES / ERP Manufacturing Operations & Quality Analytics

Business Problem & Project Objectives

The Challenge (Raw ERP Data)

Raw MES/ERP production log exports (.txt) suffer from heavy missing values (NULLs), inconsistent date formats, untrimmed strings, and invalid zero-divisors in yield calculations.

Project Goal

Design an automated SQL Server cleaning & transformation pipeline, establish a robust Star Schema, and build a Power BI executive dashboard for Yield and Material Loss tracking.

Key Business Impact: Eliminate reporting crash errors, guarantee 100% data integrity, and reduce manual reporting latency from raw text exports down to automated real-time insights.

Key Metrics Tracked

- **Overall First Pass Yield (FPY %):** $\text{Good Qty} / \text{Produced Qty}$
- **Scrap / Reject Rate (%):** $\text{Rejected Qty} / \text{Produced Qty}$
- **Material Loss / Discrepancy:** $\text{Target WO Qty} - \text{Actual Produced Qty}$

Operational Scope

Covers 67 raw log attributes across Work Orders (WO), Operations (Weaving, Cutting, Packing), Machine Centers, Shift Teams, and Product Variants.

End-to-End Pipeline Architecture

- 1 Raw Data Ingestion (.txt):** Bulk load flat ERP logs into SQL staging table Raw_ERP_Data with flexible string data types.
- 2 SQL Cleaning & Transformation:** Execute robust ETL script utilizing NULLIF, COALESCE, TRY_CAST, and zero-division protection to populate Fact_Production_Clean.
- 3 Star Schema Modeling:** Establish direct relationships between Fact_Production_Clean and a dedicated DAX Date Dimension (Dim_Date).
- 4 Power BI DAX & Analytics:** Compute dynamic measures for Yield, Scrap Rate, and Over-production alerts.

Pipeline Advantage

By performing structural transformation and date fallback logic at the SQL layer, downstream Power BI reports load faster, remain lightweight, and eliminate (Blank) visual artifacts.

SQL Data Cleaning & Business Logic

Field Category	Issue in Raw Data	SQL Server Solution & Logic
Text / Attributes	Blank strings and NULLs in Customer / Item names	ISNULL(NULLIF(TRIM([Cust Name]), ''), 'Unknown Customer')
Work Order Dates	Missing WODate breaks calendar relationships	COALESCE(TRY_CAST(WODate AS DATE), TRY_CAST(FiscalDate AS DATE))
Delivery Dates	NULLs in Make-to-Stock inventory items	Preserve NULL (Standard ERP practice for stock production)
Yield Percentage	Division by zero when ProducedQty = 0	CASE WHEN ProducedQty > 0 THEN (GoodQty / ProducedQty)*100 ELSE 0 END
Material Loss	Negative quantities due to unadjusted balance	WOQty - ProducedQty (Reflects pending production requirement)

```
-- Core Transformation SQL Snippet
COALESCE(TRY_CAST([WO Date] AS DATE),
TRY_CAST([Fiscal Date] AS DATE)) AS WODate, (ISNULL(TRY_CAST([Produced Qty] AS FLOAT), 0) -
ISNULL(TRY_CAST([Rejected Qty] AS FLOAT), 0)) AS GoodQty, CASE WHEN ISNULL(TRY_CAST([Produced
Qty] AS FLOAT), 0) > 0 THEN ROUND(((TRY_CAST([Produced Qty] AS FLOAT) -
ISNULL(TRY_CAST([Rejected Qty] AS FLOAT), 0)) / TRY_CAST([Produced Qty] AS FLOAT)) * 100, 2)
ELSE 0.0 END AS YieldPercentage
```

Star Schema & DAX Measures

Star Schema Structure

- **Fact Table:** Fact_Production_Clean
- **Dimension Table:** Dim_Date (Generated via DAX CALENDAR)
- **Active Relationship:** Dim_Date[Date] 1
—— * Fact_Production_Clean[WODate]

Measure Table Architecture

All calculations are organized in a dedicated _Measures table for central maintenance and optimized evaluation context.

Key DAX Measure Definitions:

```
// 1. Overall First Pass Yield (%) Overall Yield % = DIVIDE([Total Good Qty], [Total  
Produced Qty], 0) // 2. Scrap / Reject Rate (%) Reject Rate % = DIVIDE([Total Rejected Qty],  
[Total Produced Qty], 0) // 3. Over-Production Work Order Count Alert Over Production WO  
Count = CALCULATE( COUNTROWS(Fact_Production_Clean), Fact_Production_Clean[ProductionStatus]  
= "Over-Production" )
```

Dashboard Page 1: Progress & Yield

Header & Slicers

Global filtering by Fiscal Year-Month, Work Center Name, Shift Code, and WO Status.

Top Executive KPI Cards

- Total Target WO Qty
- Total Actual Produced Qty
- Overall Yield % (Green ≥ 98%, Red < 95%)
- Scrap Rate %

Primary Visual: Combo Chart (Work Center Performance)

- **Bar Chart (Y-Axis 1):** Total Produced Quantity by Operation (Weaving, Cut & Fold, Packing).
- **Line Chart (Y-Axis 2):** Overall Yield % per operation.
- **Business Insight:** Immediately highlights operation bottlenecks and quality drops across lines.

Detailed Matrix Visual

Hierarchical drill-down: WO Number → Item Name displaying Target Qty, Produced Qty, Good Qty, Rejected Qty, Yield %, and Material Loss Qty.

Dashboard Page 2: Compliance & Machine Alert

Over-Production Alert Visual

Filters orders flagged as Over -Production (Produced Qty > WO Target Qty) to prevent unbudgeted raw material consumption.

Cost vs Quality Correlation

Scatter plot comparing Daily Machine Cost / RPM Speed against Rejected Quantity to identify machine wear patterns.

QA/QC Operational Table: Live alert list displaying Work Orders with Scrap Rate > 3% tagged with Operator Name and Machine Code for direct supervisor intervention.

WO Number	Work Center	Operator	Status Flag	Action Required
151652694	Cut & Fold	Shruti Singh	High Reject (>3%)	Inspect blade alignment
151652701	Weaving	Rakesh	Over-Production	Review WO closure rules

Business Impact & Project Summary

100% Data Integrity

Eliminated all #DIV/0! errors, blank slicers, and negative material losses, delivering a reliable single source of truth.

Real-Time QA Insights

Enabled proactive quality monitoring, reducing defect detection time from days to automated shift-level tracking.

Project Summary & Technical Deliverables

- ✓ **SQL Server Pipeline:** Fully functional data cleaning script with date fallback and null-handling.
- ✓ **Power BI Model:** Optimized Star Schema with custom DAX date table and measure library.
- ✓ **Interactive Reporting:** 2-Page Executive Dashboard for yield tracking and QA alerts.
- ✓ **Documentation:** Comprehensive Data Dictionary and GitHub portfolio assets.

Thank You! Open for Questions & Discussion.